

Education

University of Guelph

Expected Graduation, May 2027

Bachelor of Computing in Software Engineering, Minor in Entrepreneurship

Guelph, Ontario

- **Dean's List**, Winter 2026
- **Relevant Coursework:** Operating Systems, Data Structures & Algorithms, System Design, Software Design, Object-Oriented Programming, Software Engineering.

Professional Experience

Software Engineer Intern

April 2026 – Aug 2026

The Simple Knot Inc.

Guelph, Ontario

- Developed and integrated **REST APIs** connecting **Instagram** and **Pinterest** into the platform, handling authentication, error states, and data mapping between services, expanding the platform's social media integrations.
- Owned end-to-end delivery of features on a production **iOS** and **Android** application serving **400+ clients**, using **AI-assisted tooling** for refactoring and test scaffolding while validating all output through code reviews and **CI/CD**.

Software Developer Intern

May 2024 – Aug 2024

Algoma Steel Inc.

Sault Ste. Marie, Ontario

- Built **Python** and **MySQL** backend workflows and **relational data models** aggregating real-time production KPIs into **PI Vision** dashboards, serving **400+ engineers** and accumulating **4,000+ views**.
- Wrote and optimized **SQL queries** against large operational datasets, cutting dashboard load times by **40%** and improving responsiveness for engineers monitoring plant operations.
- **Debugged** and resolved **data pipeline** inconsistencies end-to-end, coordinating with plant engineers to validate data accuracy in an environment with **zero downtime tolerance**.

Projects

AI Codebase Q&A | Python, FastAPI, PostgreSQL, Redis, RAG, pgvector, Gemini API, Next.js

Aug. 2026 – Present

- Built an async **FastAPI** pipeline that clones **GitHub** repos and chunks them by function with **tree-sitter**, using **Redis** workers to keep latency independent of repo size.
- Implemented hybrid retrieval over **pgvector**, fusing semantic and full-text rankings to assemble context for the **Gemini API**, returning answers cited to exact file and line ranges.
- Designed per-IP rate limiting with an atomic **Redis Lua** script (5 indexes/hr, 30 queries/min), capping **Gemini API** cost and preventing abuse while failing open to preserve availability.

NYC Complaint Heatmap | Spring Boot, Java, Leaflet.js, Mapbox, JUnit, NYC Open Data

Jan. 2026 – Apr. 2026

- Designed RESTful API endpoints in **Spring Boot** querying large-scale **NYC Open Data** datasets and returning **GeoJSON** to the frontend, enabling real-time heatmap rendering with **Leaflet.js** and **Mapbox**.
- Structured service and controller layers using object-oriented design patterns and wrote **JUnit** tests covering service logic and endpoint validation across all complaint categories, ensuring correctness as new data sources were added.

Skills

- **Languages:** Java, Python, JavaScript/TypeScript, SQL, C#, C.
- **Frontend:** React.js, Next.js, HTML5/CSS3, Tailwind CSS.
- **Backend & Databases:** Spring Boot, Node.js, MongoDB, FastAPI, PostgreSQL, MySQL, Redis, SQL Server.
- **AI & Emerging Tech:** LLM integration, Generative AI, RAG, Vector Search (pgvector).
- **Tools & Practices:** Git, Docker, CI/CD, AWS, Azure, Agile/Scrum, JUnit, Jest, Jira, Postman, DevOps, Software Engineering Life Cycle, Code Review, Technical Documentation, Organization, Leadership, Communication.

Leadership Experience

Director of Finance

Sept. 2025 – May 2026

Google Developer Groups

Guelph, Ontario

- Secured **\$7,200+** in sponsorships through outreach to **100+ companies**, fully funding a **250-person hackathon** delivered on time and within budget.
- Coordinated with a **10-person** organizing team across logistics, marketing, and judging to deliver the event, managing budget allocation and vendor payments end-to-end.